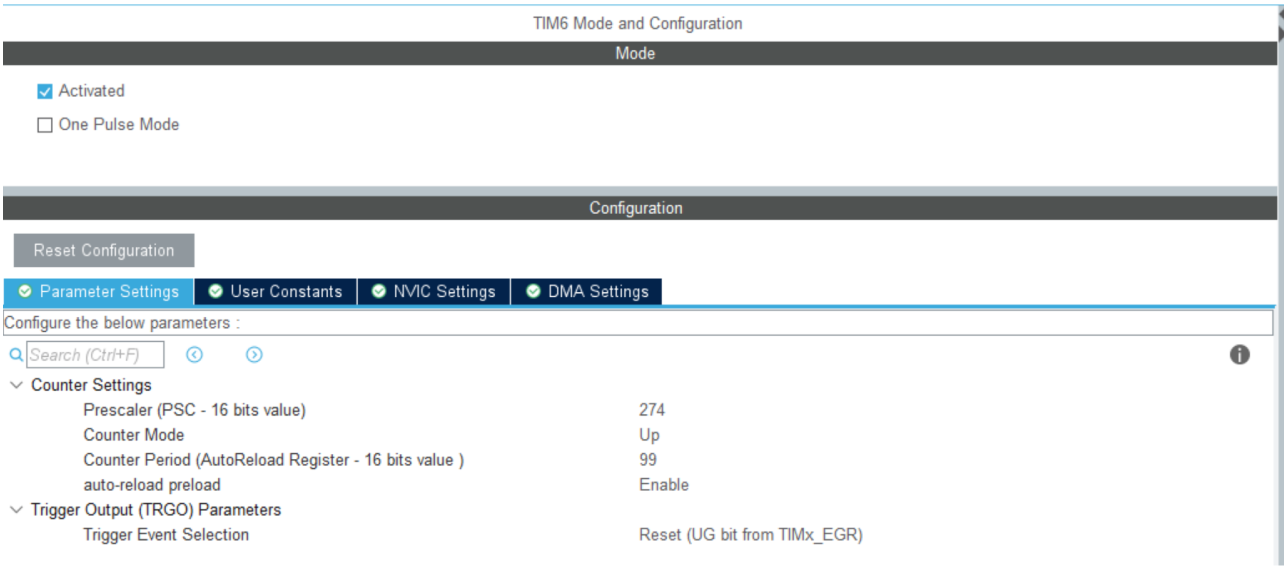
1. 文档说明

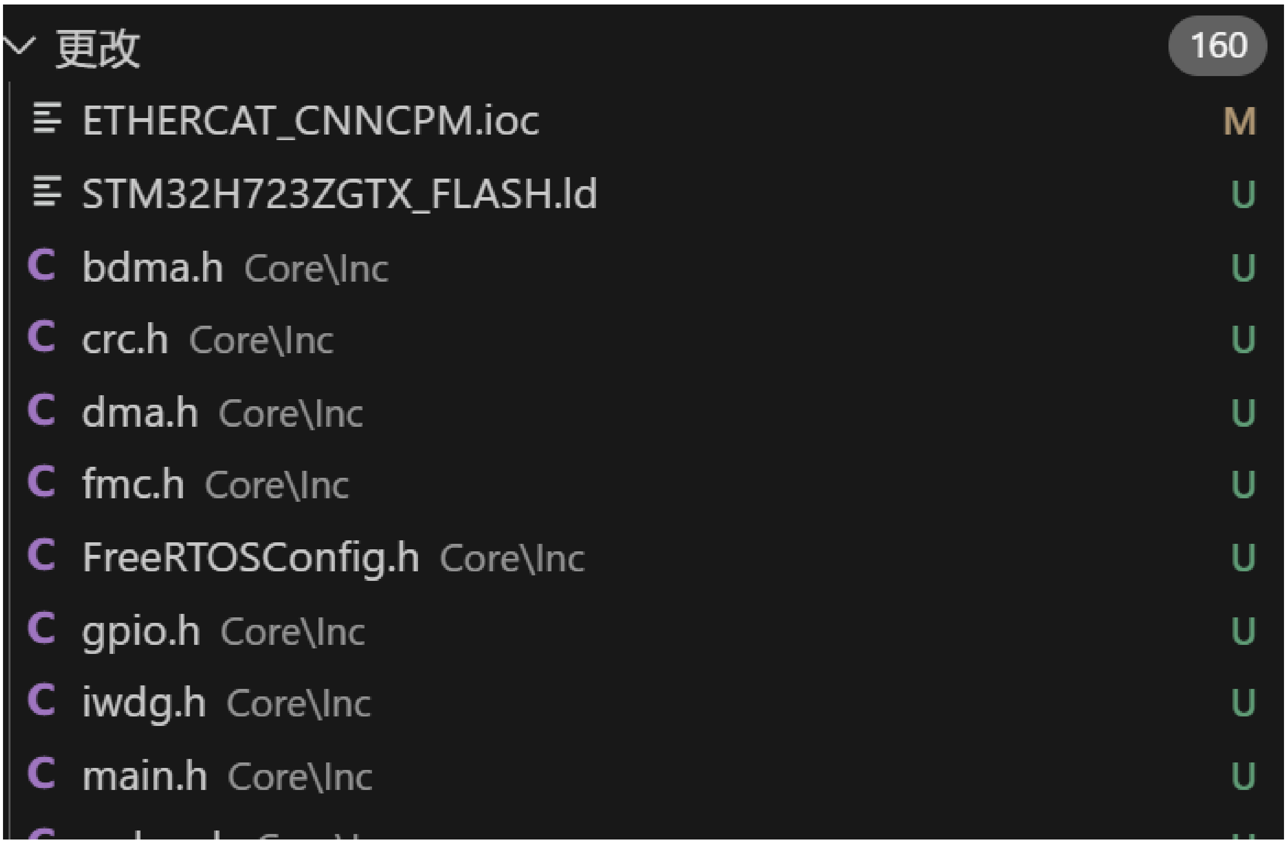
- 本demo针对新增片上外设，且不需要设备驱动情况时，工程修改参考
- 本demo拟实现如下功能：
  - 使用TIM6，中断周期为0.1ms，从而提供时间基准供FreeRTOS进行任务运行耗时统计
  - TIM6不占用GPIO，且不被业务需求依赖，故按工程拟定的原则，需要将其初始化放置到**main**函数内部
  - 基础分支hash: 8f895773ff55a79bed5662de1b436ddf827d4cd4
  - demo分支hash: 1b43d79b5e6d06a3eaa587a3097d763b7057e294

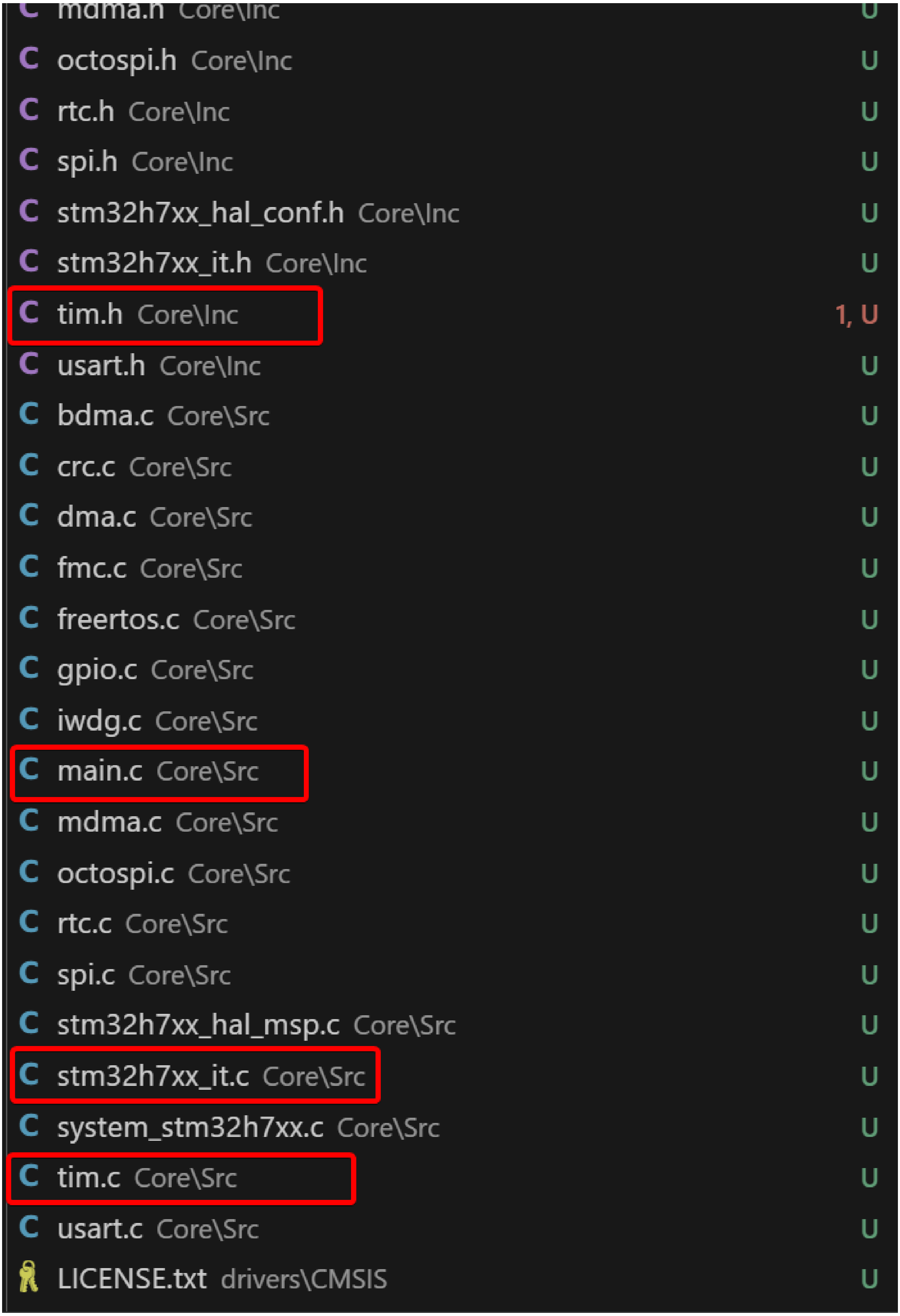
2. 修改方式

- 使用STM32CubeMX构建TIM6配置



- 生成工程，并筛选TIM6相关代码文件





- 将步骤2中文件与原文件对比，差异部分拷贝到相应的原文件中

main.c

```c
  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_DMA_Init();
  MX_MDMA_Init();
  MX_BDMA_Init();
  MX_FMC_Init();
  MX_TIM1_Init();
  MX_TIM2_Init();
  MX_CRC_Init();
  MX_IWDG1_Init();
  MX_RTC_Init();
  MX_TIM6_Init();
```

stm32h7xx_it.c

```c
/* USER CODE END 0 */

/* External variables --------------------------------------------------------*/
extern TIM_HandleTypeDef htim1;
extern TIM_HandleTypeDef htim6;
/* USER CODE BEGIN EV */
```

```c
/**
  * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
  */
void TIM6_DAC_IRQHandler(void)
{
  /* USER CODE BEGIN TIM6_DAC_IRQn 0 */

  /* USER CODE END TIM6_DAC_IRQn 0 */
  HAL_TIM_IRQHandler(&htim6);
  /* USER CODE BEGIN TIM6_DAC_IRQn 1 */

  /* USER CODE END TIM6_DAC_IRQn 1 */
}
```

tim.c

```c
TIM_HandleTypeDef htim1;
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim6;
```

```c
/* TIM6 init function */
void MX_TIM6_Init(void)
{

  /* USER CODE BEGIN TIM6_Init 0 */

  /* USER CODE END TIM6_Init 0 */

  TIM_MasterConfigTypeDef sMasterConfig = {0};

  /* USER CODE BEGIN TIM6_Init 1 */

  /* USER CODE END TIM6_Init 1 */
  htim6.Instance = TIM6;
  htim6.Init.Prescaler = 274;
  htim6.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim6.Init.Period = 99;
  htim6.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_ENABLE;
  if (HAL_TIM_Base_Init(&htim6) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim6, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM6_Init 2 */

  /* USER CODE END TIM6_Init 2 */

}
```

```c
void HAL_TIM_Base_MspInit(TIM_HandleTypeDef* tim_baseHandle)
{

  if(tim_baseHandle->Instance==TIM1) ···
  else if(tim_baseHandle->Instance==TIM2) ···
  else if(tim_baseHandle->Instance==TIM6)
  {
  /* USER CODE BEGIN TIM6_MspInit 0 */

  /* USER CODE END TIM6_MspInit 0 */
    /* TIM6 clock enable */
    __HAL_RCC_TIM6_CLK_ENABLE();

    /* TIM6 interrupt Init */
    HAL_NVIC_SetPriority(TIM6_DAC_IRQn, 5, 0);
    HAL_NVIC_EnableIRQ(TIM6_DAC_IRQn);
  /* USER CODE BEGIN TIM6_MspInit 1 */

  /* USER CODE END TIM6_MspInit 1 */
  }       You, 27秒钟前 • Uncommitted changes
}
```

```c
void HAL_TIM_Base_MspDeInit(TIM_HandleTypeDef* tim_baseHandle)
{

  if(tim_baseHandle->Instance==TIM1) ...
  else if(tim_baseHandle->Instance==TIM2) ...
  else if(tim_baseHandle->Instance==TIM6)
  {
  /* USER CODE BEGIN TIM6_MspDeInit 0 */

  /* USER CODE END TIM6_MspDeInit 0 */
    /* Peripheral clock disable */
    __HAL_RCC_TIM6_CLK_DISABLE();

    /* TIM6 interrupt Deinit */
    HAL_NVIC_DisableIRQ(TIM6_DAC_IRQn);
  /* USER CODE BEGIN TIM6_MspDeInit 1 */
      You, 55秒钟前 • Uncommitted changes
  /* USER CODE END TIM6_MspDeInit 1 */
  }
}
```

tim.h

```c
extern TIM_HandleTypeDef htim1;

extern TIM_HandleTypeDef htim2;

extern TIM_HandleTypeDef htim6;

/* USER CODE BEGIN Private defines */

/* USER CODE END Private defines */

void MX_TIM1_Init(void);
void MX_TIM2_Init(void);
void MX_TIM6_Init(void);
```

- 重新加载运行cmakelist.txt，以更新makefile文件

- 编译文件，结果如下

```
问题 5   输出   调试控制台   终端   端口   MEMORY   XRTOS   GITLENS

[ 86%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_pwr_ex.c.obj
[ 87%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc.c.obj
[ 88%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc_ex.c.obj
[ 89%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc.c.obj
[ 90%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc_ex.c.obj
[ 91%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_sdram.c.obj
[ 92%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi.c.obj
[ 93%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi_ex.c.obj
[ 94%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim_ex.c.obj
[ 96%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart.c.obj
[ 97%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart_ex.c.obj
[ 98%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_ll_fmc.c.obj
[100%] Linking C executable ETHERCAT_CNNCPM.elf
Memory region         Used Size  Region Size  %age Used
         ITCMRAM:          0 GB        64 KB      0.00%
         DTCMRAM:          0 GB       128 KB      0.00%
           FLASH:      161440 B         1 MB     15.40%
          RAM_D1:      243704 B       320 KB     74.37%
          RAM_D2:          1 KB        32 KB      3.13%
          RAM_D3:         266 B        16 KB      1.62%
         RAM_BKP:           6 B         4 KB      0.15%
Building F:/WORK/arm_core_develop/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.hex
Building F:/WORK/arm_core_develop/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.bin
[100%] Built target ETHERCAT_CNNCPM.elf
PS F:\WORK\arm_core_develop\FW-ArmCore\ArmCoreV1_QSPI\build>
```

- 添加代码：启动TIM6，并于中断中添加打印测试

```c
/* TIM6 init function */
void MX_TIM6_Init(void)
{

  /* USER CODE BEGIN TIM6_Init 0 */

  /* USER CODE END TIM6_Init 0 */

  TIM_MasterConfigTypeDef sMasterConfig = {0};

  /* USER CODE BEGIN TIM6_Init 1 */

  /* USER CODE END TIM6_Init 1 */
  htim6.Instance = TIM6;
  htim6.Init.Prescaler = 274;
  htim6.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim6.Init.Period = 99;
  htim6.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_ENABLE;
  if (HAL_TIM_Base_Init(&htim6) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim6, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM6_Init 2 */
  HAL_TIM_Base_Start_IT(&htim6);
  /* USER CODE END TIM6_Init 2 */

}
```

```c
/**
  * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
  */
void TIM6_DAC_IRQHandler(void)
{
  /* USER CODE BEGIN TIM6_DAC_IRQn 0 */
  printf("tim6 running\r\n");
  /* USER CODE END TIM6_DAC_IRQn 0 */
  HAL_TIM_IRQHandler(&htim6);
  /* USER CODE BEGIN TIM6_DAC_IRQn 1 */

  /* USER CODE END TIM6_DAC_IRQn 1 */
}
```

- 结果如下

```
[17:27:25.580]收←◆
************************************
fw version: 00.00.01
compile time: Jan 31 2024  17:14:05
git branch: arm_core_master_restructure_temporary
git hash: 8f895773ff55a79bed5662de1b436ddf827d4cd4
mcu clock:550.00 M
stm32 uid:0x3836353732315118001000 1e
flash size:1024KB
hal driver version:01.11.01.00
************************************

########## function initialize begin ##########
done_1   [backup_ram_clk_enable          10]
done_2   [fram_init                       0 ]
done_3   [gpio_port_init                  0 ]
done_4   [component_cm_backtrace_init     0 ]
done_5   [component_ulog_init             0 ]
done_6   [ethercat_thread_init            0 ]
done_7   [fpga_thread_init                0 ]
done_8   [non_realtime_process_thread_init 0 ]
done_9   [tcp_client_thread_init          0 ]
done_10  [ulog_thread_init                0 ]
done_11  [console_thread_init             0 ]
########## function initialize end   ##########

tim6 running
tim6 running
tim6 running
tim6 running
[im6 running
:00:00.000] [INFO]: Init ok
rim6 running
 reg (0x64), readData tim6 running
is: 0x87654321
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
tim6 running
```

- 最后，添加FreeRTOS统计功能函数及实现函数即可

```c
/**
  * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
  */
void TIM6_DAC_IRQHandler(void)
{
  /* USER CODE BEGIN TIM6_DAC_IRQn 0 */

  /* USER CODE END TIM6_DAC_IRQn 0 */
  HAL_TIM_IRQHandler(&htim6);
  /* USER CODE BEGIN TIM6_DAC_IRQn 1 */
#ifdef configGENERATE_RUN_TIME_STATS
  run_time_count_increase();
#endif
  /* USER CODE END TIM6_DAC_IRQn 1 */
}
```